

1. Hypertension in adults (incl. T2DM)

In women trying to conceive/pregnant/breastfeeding: follow NICE guidance on hypertension in pregnancy
Based on NICE NG136, 2019

DIAGNOSIS OF HYPERTENSION

- Use an appropriately-sized cuff.
- If clinic BP $\geq 140/90$, repeat. If 2nd BP substantially different from 1st, do 3rd BP: record **LOWEST** of last 2 BP.
- If pulse is irregular, automated machines may not measure BP accurately – auscultate manually.
- When diagnosing hypertension, measure BP in both arms: if difference $>15\text{mmHg}$, repeat. If difference remains $>15\text{mmHg}$, measure all subsequent readings in higher arm.

Use ABPM for DIAGNOSIS. Use CLINIC readings for SCREENING/ASSESSING RESPONSE TO TREATMENT

Clinic BP **$<140/90$**

Clinic BP **$140-179$
 $90-119$**

Clinic BP **$\geq 180/120$ = severe hypertension**

NORMOTENSIVE

Recheck BP at least every 5y, more frequently if close to 140/90

Same-day referral if:

- **Accelerated hypertension:** papilloedema/retinal haemorrhages.
- **Suspected pheochromocytoma:** labile BP (hypertension, postural hypotension), headache, palpitations, pallor, excessive sweating.
- **Life-threatening features:** confusion, chest pain, heart failure, AKI.

If asymptomatic and none of above: look for target organ damage ASAP (see below):

- **Target organ damage:** treat immediately without ABPM/HBPM.
- **No target organ damage:** repeat clinic BP within 7d. Work up/diagnose/treat as appropriate.

INVESTIGATIONS: use the results to decide on treatment pathway (see overleaf)

1. Offer ambulatory monitoring (ABPM) Home readings if unable to tolerate ABPM	2. Look for target organ damage (damage to CV system, kidney, eyes, e.g. LVH)	3. Assess CV risk
ABPM: 2 readings/h in waking hours only*. You need at least 14 readings to average. HBPM: 2 readings twice daily for at least 4d, preferably 7d. Discard first day's readings and average ALL the other readings. *You therefore won't spot night-time non-dippers (who fail to drop their BP at night and who are at increased CVD risk). NICE does not discuss this: more info in main article.	• ECG. • Urine: ACR and dip urine for haematuria. • Bloods: U&Es, HbA1c, cholesterol and HDL. • Fundoscopy: for hypertensive retinopathy. If target organ damage found but ABPM is normal, look for alternative (non-BP) cause, e.g. heart failure, CKD.	Use formal tool, e.g. latest QRISK.

REFERRAL CRITERIA

Same-day	• Accelerated hypertension or suspected pheochromocytoma (see above for features).
Routine	• Consider in those under 40y with hypertension: need a more detailed assessment for target organ damage, and because 10y CVD risk estimations tend to underestimate lifetime risk. • Consider if secondary hypertension suspected. • Unresponsive on step 4 therapy.

POSTURAL BP DROP (new and established hypertensives): check BP lying (seated if inconvenient to lie down), then after standing for at least 1 minute. Significant drop is **SYSTOLIC** drop $\geq 20\text{mmHg}$ OR **DIASTOLIC** drop $\geq 10\text{mmHg}$. If postural symptoms and seated-standing BP doesn't show a significant drop, check lying-standing BP. Important to spot because increased risk of overtreatment and resultant risk of falls and associated harms.

Look for postural BP drop if: • $\geq 80\text{y}$. • Type 2 diabetes. • Postural symptoms (falls/postural dizziness).	If <u>SYSTOLIC</u> BP drops $\geq 20\text{mmHg}$ <u>or</u> <u>DIASTOLIC</u> BP drops $\geq 10\text{mmHg}$: • Review likely causes, including drugs. Manage appropriately. If postural symptoms and these persist despite addressing causes, consider referral. • Use <u>STANDING</u> BP for future BP recordings, and aim to get <u>STANDING</u> BP (not sitting BP) below treatment targets.
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We make every effort to ensure the information in these pages is accurate and correct at the date of publication, but it is of necessity of a brief and general nature. The information presented herein should not replace your own good clinical judgement, or be regarded as a substitute for taking professional advice in appropriate circumstances. In particular, we suggest you carefully consider the specific facts, circumstances and medical history of any patient, and recommendations of the relevant regulatory authorities. We also suggest that you check drug doses, potential side-effects and interactions with the British National Formulary. Save insofar as any such liability cannot be excluded at law, we do not accept any liability for loss of any type caused by reliance on the information in these pages. May 2024. For full references see the relevant Red Whale articles.

2. Management of hypertension

(including T2DM, but NOT those with CKD or women who are pregnant/trying to conceive/breastfeeding)

Based on NICE 2019, NG136

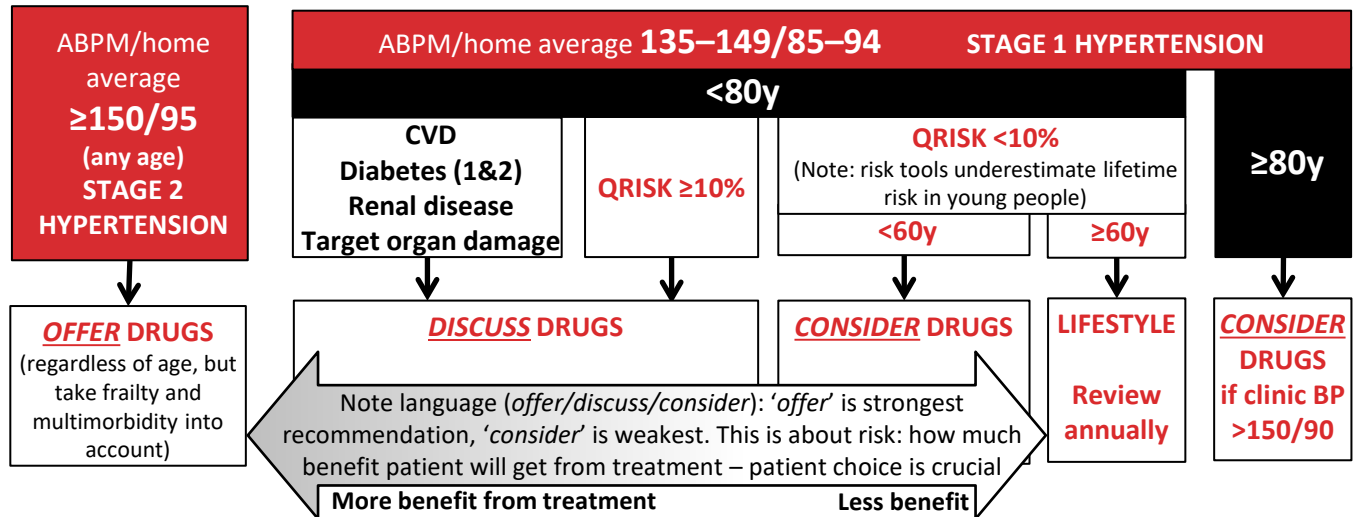
Lifestyle!



LIFESTYLE FOR EVERYONE!

- Smoking cessation
- Weight loss if overweight (1kg loss = 1mmHg SBP drop) (JAMA 2022;328:1849)
- Regular physical activity and healthy diet

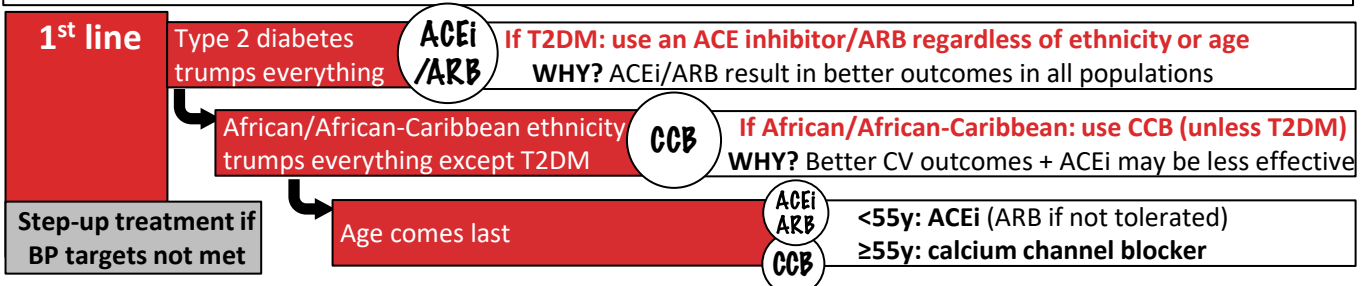
If <40y with hypertension: **CONSIDER REFERRAL** (to look for cause/assess pros/cons of treatment)



Which drug to start and in which order? (use the following unless contraindicated)

Lifestyle advice still important for those on drugs! Bear in mind frailty, multimorbidity and comorbidities.

- If **CVD**, follow NICE guidance for that disease. If still hypertensive, follow guidance below. In **heart failure**, use indapamide and follow NICE heart failure guidance (may include using ACEi/ARB + cardioselective beta-blocker).
- If **trying to conceive/pregnant/breastfeeding**, follow NICE hypertension in pregnancy guidance.
- If **African/African-Caribbean** and an ACEi/ARB is indicated, consider ARB (less risk of angioedema than with ACEi).



2nd line Any 2 from: ACEi/ARB, CCB, thiazide-like diuretic (indapamide not bendroflumethiazide)

3rd line ACEi/ARB + CCB + thiazide-like diuretic

Before 4th drug: check compliance/confirm BP elevated with ABPM or home average/check for postural drop
Irish study found almost half didn't have resistant hypertension when compliance and ABPM checked (BJGP 2019; 69:e621)

4th line = resistant hypertension

- Potassium ≤ 4.5 : add spironolactone 25mg/d (unlicensed). Risk of hyperkalaemia, especially if low eGFR: check U&Es within 1m and regularly thereafter (monitoring schedule in AKI article).
- Potassium >4.5: consider adding alpha/beta-blocker.
- If BP not controlled on 4 drugs, REFER.

TARGETS on TREATMENT (adapt for age/frailty)		Clinic	ABPM/home average
Remember to look for postural drop if $\geq 80y$ or T2DM or postural symptoms			
• Monitoring: use clinic readings unless white coat/masked hypertension. (White coat: >20/10 difference between average ABPM/home readings and clinic BP at diagnosis. Masked hypertension: ABPM/home readings HIGHER than clinic readings.) • Review patients annually: step-up if targets not met.	<80y	<140/90	<135/85
	$\geq 80y$	<150/90	<145/85
	If CKD (any age), clinic target based on ACR		
	ACR <70	<140/90	
NB: if type 1 diabetes:		ACR ≥ 70	<130/80
• If <80y AND ACR ≥ 70, clinic target is <130/80 • If $\geq 80y$, target is <150/90 regardless of ACR			